Experiment 35: Exothermic and endothermic processes

Notes

Chemical reactions are accompanied by the release or absorption of energy. Reactions in which energy is released are described as **exothermic**. Those in which energy is absorbed are **endothermic**. The energy released in chemical reactions was previously stored as chemical potential energy in the reactants. This stored chemical potential energy is called the heat content or *enthalpy* of a substance.

In this experiment you will be observing and classifying chemical processes as endothermic or exothermic based on the temperature changes you observe.

Equipment

thermometer (-10 to 110 °C)
100 mL beaker
plastic teaspoon
steel wool
distilled water
saturated copper(II) sulfate solution [$CuSO_4$] (25 mL)
The following solids (about a teaspoonful of each):

- sodium hydroxide[NaOH]
- sodium acetate [$NaCH_3COO$]
- barium hydroxide [$Ba(OH)_2$]
- ammonium chloride [$NH_4C\ell$]
- sodium chloride [$NaC\ell$]
- ammonium thiocyanate [NH_4SCN]

SAFETY NOTE:
- **Sodium hydroxide and barium hydroxide are corrosive and must not be allowed to come in contact with your skin.**

Procedure

Part A: solution processes

1. Place about 30 mL of water in a 100 mL beaker and measure its temperature. Add one teaspoonful of sodium hydroxide pellets, stir gently, and record any change in the temperature.
2. Repeat the procedure using, in turn, a teaspoonful of ammonium chloride, sodium acetate and sodium chloride.
3. Tabulate your results under the headings: solute, initial temperature, final temperature, and classify the reaction as exothermic or endothermic.

Procedure

Part B: reaction between ammonium thiocyanate and barium hydroxide

1. Place about half a teaspoonful each of solid ammonium thiocyanate and solid barium hydroxide in a test tube. Cork the test tube and shake gently until a solution is formed.
2. Remove the cork and smell cautiously. Try to identify one of the products. Measure and record the temperature of the solution.

Procedure

Part C: reaction between iron and copper(II) sulfate solution

1. Place about 25 mL of saturated copper(II) sulfate into a 100 mL beaker and record the temperature.
2. Place a cylinder of steel wool about 1 cm thick into the solution and hold it under the surface with the thermometer. Record any evidence of reaction and change in temperature.

Processing of results and questions

Write equations for the processes or reactions in parts A, B and C and classify each reaction as either exothermic or endothermic.